William Bowley

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Summary

Electrical Engineering student specializing in electromagnetic systems, power electronics, and numerical tool development. Experienced in developing power electronics and sensor PCBs, electric motors and actuators, simulation models, and engineering software.

Technical Skills

- **Programming:** Python, C/C++, JavaScript, MATLAB
- **Simulation & CAD:** FEMM, LTspice, Fusion 360, Inventor, Altium, KiCad, PSIM (Basics)
- **Embedded & Hardware:** STM32, PCB Design, Motor Design/Control, CAN, Sensor Integration
- **Software:** Git, Linux, LaTeX/Markdown, FastAPI/Flask, React (Basics)

Projects

OpenLSM *Aug 2025 – Present*
Electrical Lead & Developer — Open Source Linear Synchronous Motor Project

- Designed, built and validated a planar linear synchronous motor prototype with a three-phase bridge driver, demonstrating closed-loop linear motion.
- Designed, built and validated a data collection board driven by an STM32 with 8 NTC thermistors, 10 μm encoder and 3-axis SPI accelerometer over RS-485.
- Designed and built a reduced-order model using the Biot–Savart law for coil slots and an axisymmetric approximation for magnets under FOC control of linear motors.
- Designed an STM32-driven 96 V, 25 A three-phase motor driver with current sensing and galvanic isolation, driven by a custom FOC implementation.
- Designed a tubular linear synchronous motor with passive cooling and custom CNC parts.

Bowley Systems *Dec 2024 – Present*
Open-source computational tools for physical systems

PyFEA — Project Lead

- Developed an intermediate representation system for multi-physics problems using solver adaptors to interface with different numerical solvers.
- Implemented the UnitValues domain-specific language, with `.uiv` and `.ut` extensions for typed numerical quantities.
- Wrote a language specification for UnitValues and published a Visual Studio Code extension for syntax highlighting.
- Alpha development available on PyPI under PyFEA with plans for public launch.

PicoUnits — Project Lead

- Developed, built and tested a runtime type system for numerical quantities.
- Built automated tests covering 70% of the UnitValues parser and numerical core.
- Achieved a 96% A rating for branching complexity across the type system and parser.
- Version 1.10 is available on PyPI under PicoUnits.

Experience

Electrics Engineer *Aug 2026 – Present*

RMIT Motorsport — Elective (LV-Isolated-DC-DC)

- Defined the implementation scope and system architecture, including converter topology selection.
- Developed and validated a First Harmonic Approximation (FHA) model for LLC resonant converters.
- Designing a 400 V to 12 V, 20 A LLC resonant converter and developing an EMI testing methodology with consideration for electrical safety.

Powertrain Engineer *Apr 2026 – Aug 2026*

RMIT Motorsport — Formula SAE Team (dyno-boards)

- Simulated Permanent Magnet Synchronous Motors (PMSMs) in PSIM for drivetrain analysis.
- Designed isolated dyno communication electronics and supported high-voltage system integration.
- Developed an analysis framework for active vs. passive precharge systems.

Hercules Challenge — IsoPod Project *Mar 2026 – Jul 2026*

RMIT University — Finalist, Commendation Award (Isopod)

- Co-designed an assistive axial shake generator for a medication reminder system (0.45 W, 8.8 Hz).
- Utilized PyFEA for electromagnetic design, achieving accuracy within 12–22% of measured results.
- Worked in a multidisciplinary team. Project reached the finals and received a Commendation Award.

Education

Bachelor of Engineering (Honours) — Electrical *Mar 2026 – Nov 2029*

RMIT University, Melbourne

Victorian Certificate of Education (VCE) *2025*

Auburn High School

Relevant studies: Mathematical Methods and Specialist, Physics, and Software Development

Additional

- Competitive swimmer — 4 × state qualifier and 7 × regional qualifier in 50 m, 100 m, and 200 m Backstroke, and 200 m Freestyle Relay.
- Community volunteering — Supported primary school fundraising events through Bunnings sausage sizzles and constructed approximately 12 garden beds for Camberwell South Primary School.